

MANUFACTURING PROCESSES-I**841**

Time Allowed: 2.5 Hours

Full Marks: 60

Answer to Question No. 1 of Group A is compulsory and to be answered first. This answer is to be made in separate loose script(s) provided for the purpose. Maximum time allowed is 30 minutes, after which the loose answer scripts will be collected and fresh answer scripts for answering the remaining part of the question will be provided. On early submission of answer scripts of Question No. 1, a student will get the remaining script earlier. Answer any Five (05) Questions from Group B.

Group A

1. Choose the correct answer from the given alternatives (any twenty):

1x20

- i. Which of the following processes is mainly used for making the connecting rod?
a) Casting b) Deep drawing c) Rolling d) Forging
- ii. The hot working of metals is carried out at _____.
a) The recrystallisation temperature b) Below the recrystallisation temperature c) Above the recrystallisation temperature d) Any temperature
- iii. Which of the following is a type of extrusion process?
a) Direct b) Indirect c) Impact d) All of the mentioned
- iv. Long wires are made by _____ process.
a) Forging b) Rolling c) Piercing d) Drawing
- v. A moving mandrel is used in _____.
a) Wire drawing b) Tube drawing c) Metal cutting d) Casting
- vi. A two high rolling mill consists of two rolls which rotate _____.
a) At the same speed and in the same direction b) At the same speed but opposite direction c) At different speeds and in the same direction d) At the different speeds and in the opposite direction
- vii. In press, which of the following mechanism / system may be used for applying power to ram?
a) Rack and pinion b) Pneumatic c) Hydraulic d) All of these
- viii. In which sheet metal operation, the punch out portion of the sheet metal is the useful part?
a) Blanking b) Piercing c) Coining d) Embossing
- ix. Which of the following part is used for holding the metal sheet during blanking operation?
a) Spherical steel ball b) Roller c) Pressure pad d) Magnet
- x. Which type of chip is formed during machining of soft and ductile material?
a) Continuous chip b) Discontinuous chip c) continuous chip with built-up edge d) None of these
- xi. How many forces are acting during orthogonal cutting action?
a) One b) Two c) Three d) Four
- xii. Which type of standard taper is being provided on the shank of Lathe centers?
a) Morse taper b) Jarno taper c) Brown and Sharpe taper d) None of these
- xiii. Pitch of the lead screw of a metric Lathe is _____.
a) 4 mm b) 6 mm c) 8 mm d) 10 mm
- xiv. The size of a Lathe is specified by the _____.
a) Length between centers b) Swing diameter over the bed c) Swing diameter over the carriage d) All of these
- xv. To perform the facing operation in a Lathe, which type of feed has to be given on cutting tool?
a) Longitudinal feed b) Cross feed c) Inclined feed d) None of these
- xvi. Which specific gear is required in change gear for cutting metric thread in an English Lathe?
a) Gear of 20 teeth b) Gear of 40 teeth c) Gear of 80 teeth d) Gear of 127 teeth

- xvii. During machining, cutting fluids are used for:
a) Cool the cutting tool b) Wash away the chips c) Cool the workpiece d) All of these
- xviii. In which Lathe operation, the effective diameter of the job get increased?
a) Turning b) Facing c) Grooving d) Knurling
- xix. Which allowance is incorporated in pattern making to compensate the shrinkage of metal during solidification?
a) Machining allowance b) Distortion allowance c) Draft allowance d) Shrinkage allowance
- xx. In order to produce hollow interior of the casting _____ are used.
a) Riser pins b) Cores c) Vent tubes d) None of these
- xxi. "Master pattern" and "lost pattern" are used in _____ casting process.
a) Continuous b) Centrifugal c) Investment d) None of these
- xxii. In gas welding, how many types of flames are used for welding?
a) One b) Two c) Three d) Four
- xxiii. In which gas welding flame three distinct cones are obtained?
a) Oxidizing flame b) Carburizing flame c) Neutral flame d) None of these
- xxiv. In TIG welding, tungsten acts as _____ electrode.
a) Consumable b) Non-consumable c) Filler d) None of these
- xxv. Which welding process requires filler material?
a) Metal arc welding b) MIG welding c) TIG welding d) Resistance welding

Group B

- 2.a. What is hot forging and cold forging?
b. With suitable diagram explain the open die forging process. 2+6
- 3.a. State the major advantages of forging process.
b. With suitable diagram explain the indirect extrusion process. 2+6
- 4.a. State the difference between orthogonal and oblique cutting.
b. State the desirable properties of cutting tool materials. 4+4
- 5.a. Define longitudinal, cross and inclined feed in respect of Lathe.
b. With suitable diagram show the nomenclature of a single point cutting tool. 3+5
- 6.a. How will you specify a Lathe?
b. State the difference between steady rest and follower rest. 4+4
- 7.a. Define odd thread and even thread.
b. Calculate the teeth number of gears and draw the gear arrangement of the change gear for cutting a screw having 7 mm pitch on a Lathe having lead screw of 4 threads per inch. 2+6
- 8.a. State the name of allowances generally being incorporated in pattern making.
b. With suitable diagram explain the centrifugal casting process. 2+6
- 9.a. State the advantages and disadvantages of welding process.
b. State the difference between gas welding and arc welding. 4+4
- 10.a. State the function of flux on a coated electrode used in arc welding.
b. With suitable diagram explain the principle of submerged arc welding. 3+5